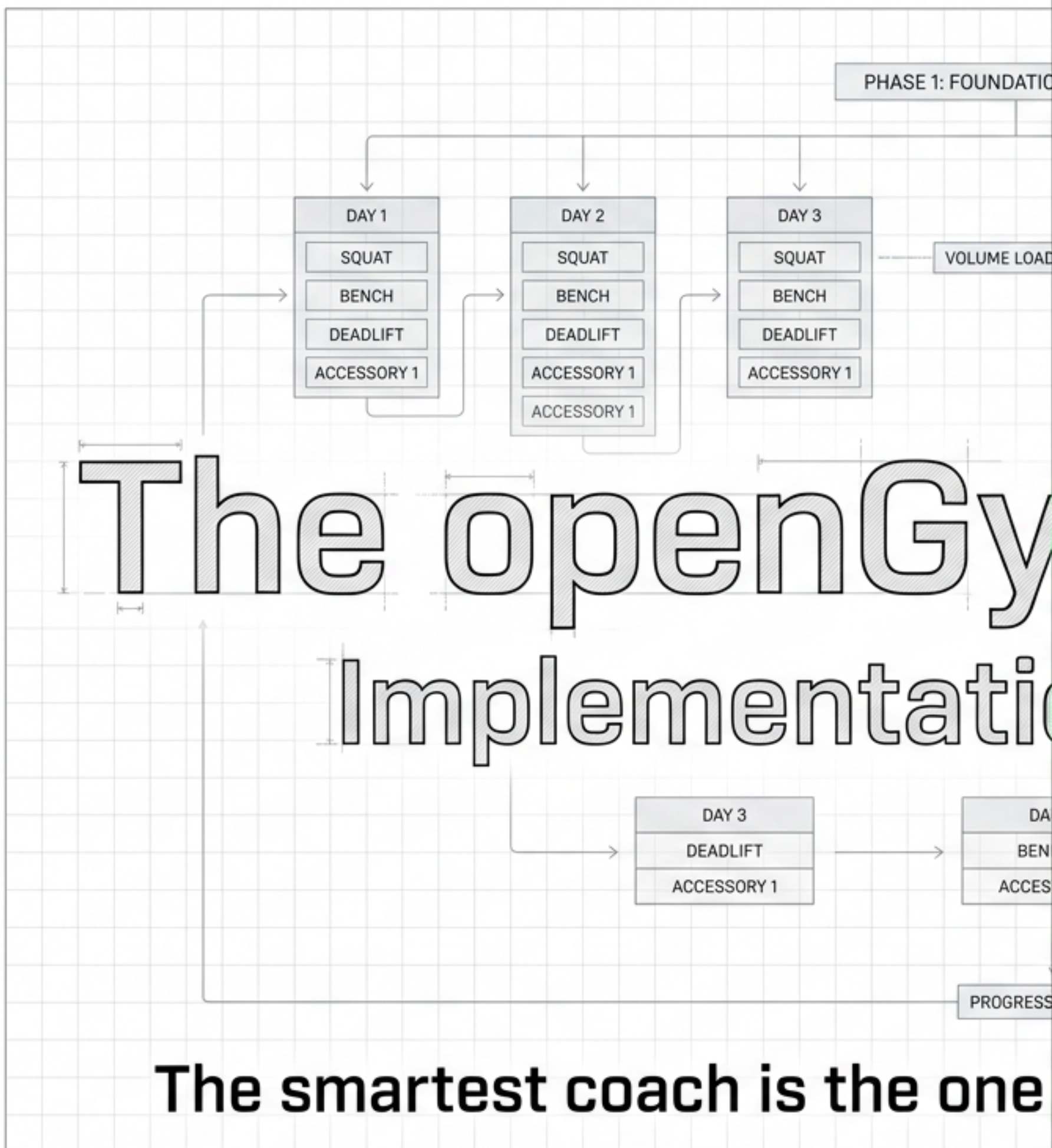




```
$ open6ym_AI --train data.py && load_model --weights v1.3.0_weights.h5 > logs/training.log  
> █
```

The openGym AI Coach

Implementation Plan v1.3.0

The smartest coach is the one running on your own machine.

SELECT STARTER PPL
(Push/Pull/Legs) from
the 1,324-exercise
library.



DAY 1

DAY 15

DAY 50

DAY 100

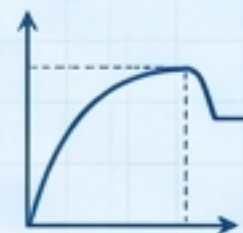
The progression engine
correctly computes
next-session loads.



LOAD:
+2.5%

openGym takes you from a starter plan
to 100 logged sessions flawlessly.
The progression engine is bulletproof.
But it cannot design.

First stall detected; the
engine fires a deload.



DELOAD
PROTOCOL
↓
-10% LOAD

100th session logged
flawlessly.



SESSION COUNT:
100/100



Drawn By:	
Checked By:	
DATE:	
Revised By:	OG-TL-001

> Adding a local, CLI-driven AI agent that designs your plan and recalculates it based on feedback—without your data ever leaving your server.

Terminal

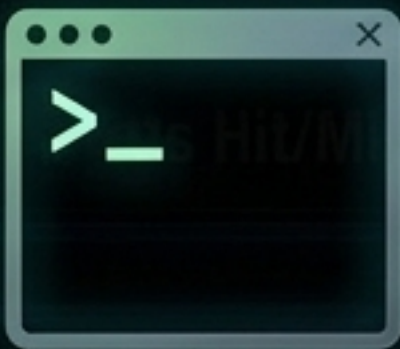
[Provider Agnostic]
Claude / Gemini / Codex.

Terminal

[Locally Hosted]
Instance owner owns
the API keys.

Terminal

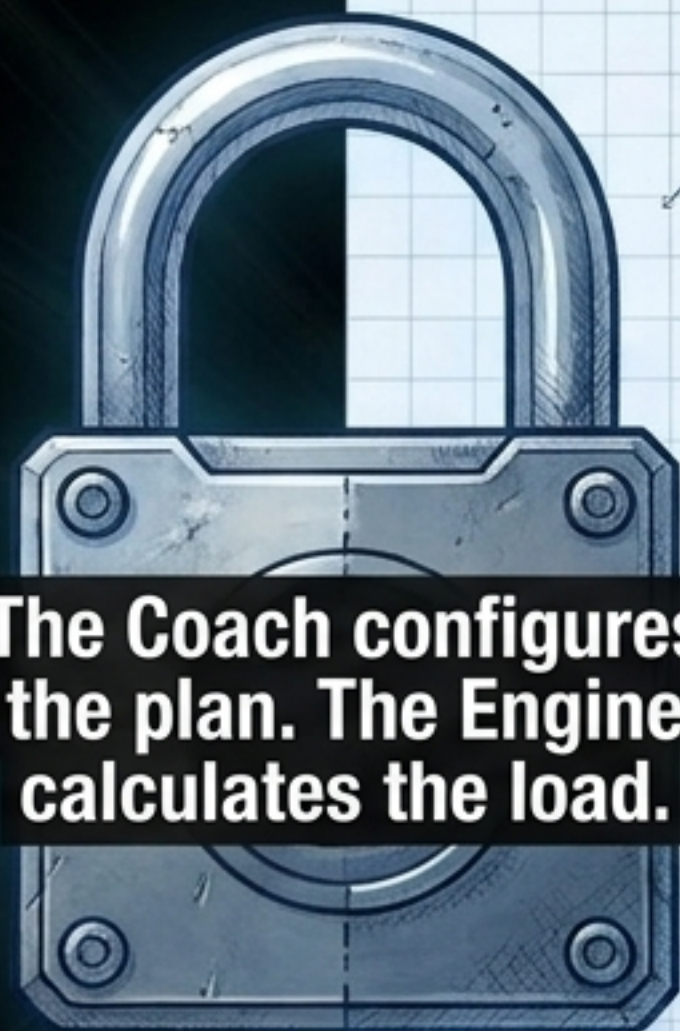
[Zero Lock-in]
App logic runs completely
independent of the cloud.



The Coach: Where Judgement Lives

- Creates weekly routines
- Sets progression policies
- Reads RPE/RIR
- Interprets plain-text feedback
- Adds/swaps exercises

The Coach configures the plan. The Engine calculates the load.



The Engine: Where Math Lives

- Computes tomorrow's bar weight
- Strictly deterministic
- Applies linear/greyskull progression
- Tracks strict hit/misses
- Manages exWeights



The 7 Principles of Implementation

Your Box, Your Data

Local CLI; openGym gains no bundled API keys.

Opt-in Twice

Admin enables it; Lifter consents to data sharing.

Auditable Trust

Every proposal is discrete, explained, and logged.

Math Stays Offline

AI doesn't dictate tomorrow's weight.

Provider-Agnostic

Claude, Gemini, Codex, or custom CLI.

Degrade Gracefully

Without AI, the app behaves exactly like today.

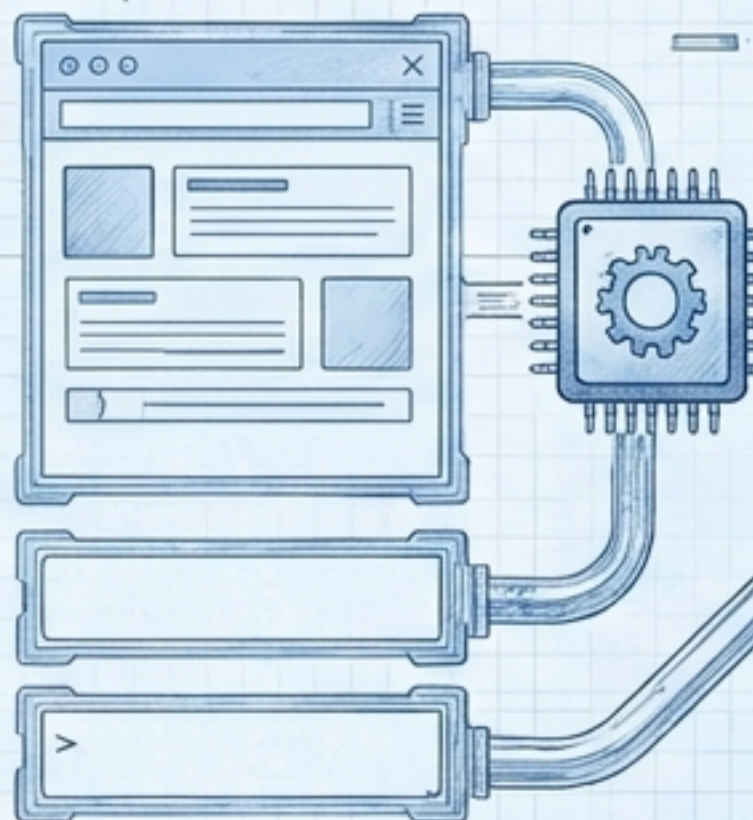
Approval Required

Proposals are inert until the user hits Apply.

Instance Host (./data)

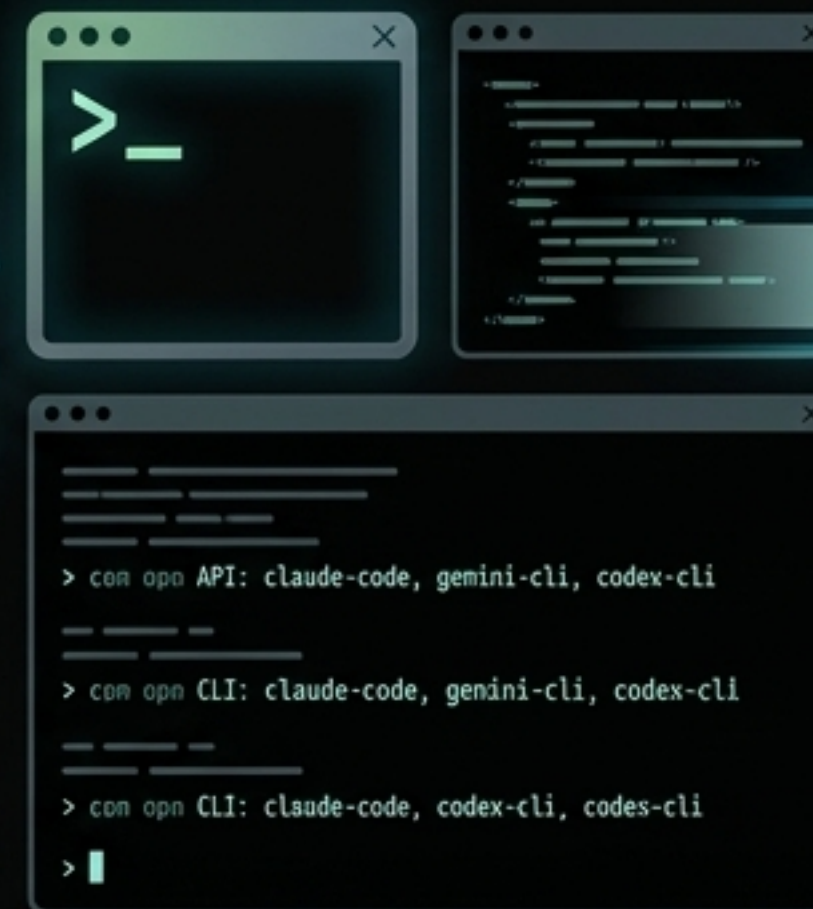


Core openGym App:
UI, Progression Engine,
State Sync



Controlled via .env
by the Instance Owner.
Core app makes
local calls only.

Headless CLI:
claude-code, gemini-cli,
codex-cli



Owner's API keys
to Model Provider
(Anthropic/
Google/OpenAI)
- completely
bypassing
openGym central
servers.

Persona Boundary Matrix

	The Instance Admin	The Lifter	The Skeptic (Non-AI User)
Pays for / provisions CLI	✓ (Yes)	✗ (No)	✗ (No)
Triggers plan recalculation	✗ (No)	✓ (Yes)	✗ (No)
Reads generated logs/prompts	✗ (No - job counts only)	✓ (Yes)	✗ (No)
App experience without consent	(Sees system status)	(Sees 'Meet the Coach' card)	(Identical byte-for-byte to v1.2.3)

Admins pay for the compute, but cannot read your prompts. Skeptics lose nothing.

Journey 1: Plan Creation

Guided Intake

Goal: Muscle

Days: 3

Equipment: Dumbbells

Limitations: Noisy neighbors

AI Proposal

Full body A/B/C

Supersets
mapped



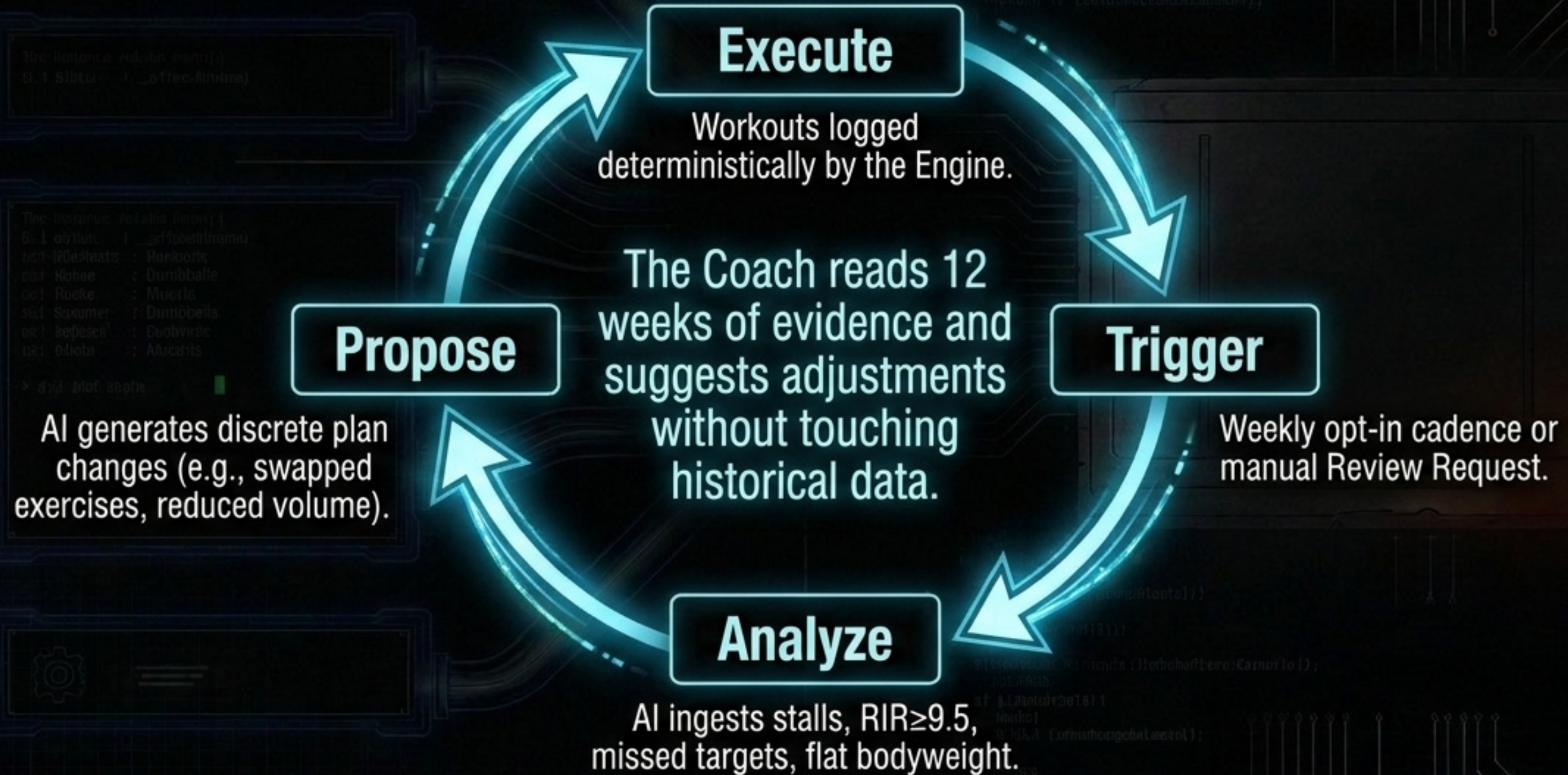
```
> Swap goblet squats for split squats.
```

Accept

Apply

Behaves identically
to importing a
.opengym_plan file.

Journey 2: The Feedback Loop



The Anatomy of Trust (The Reversible Diff)

The Evidence:
Based on your last 9 sessions (4 weeks).

Granular Control:
Per-change
Accept/Reject toggles.

Plan ProposalHistorySettings

Proposal based on evidence from last 9 sessions (4 weeks)

Exercise	Current Plan (Before)	Proposed Plan (After)	Rationale	Action
Bench Press	5x5 (Red)	Dumbbell Press 3x10 (Green)	Bench pressed at RPE ≥ 9.5 for three sessions and stalled twice. Switching to Dumbbell Press for hypertrophy focus and reduced CNS load.	☒ Accept ☐ Reject
Squat	Squat 3x8	Leg Press 4x12	Bench pressed at RPE ≥ 9.5 for three sessions and stalled twice. Switching to Dumbbell Press for hypertrophy focus and reduced CNS load.	☐ Accept ☐ Reject
Presses	Squat	Leg Press	Bench pressed at RPE ≥ 9.5 for three sessions and stalled twice. Switching to Dumbbell Press for	☐ Accept

Apply & Snapshot (One-tap Revert)

Discrete Changes:
Line-by-line diffs with specific rationales.
“Bench pressed at RPE ≥ 9.5 for three sessions and stalled twice.”

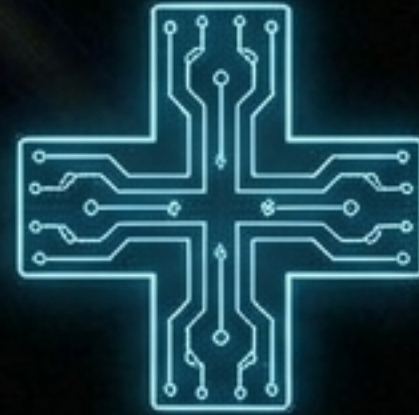
The Safety Net:
The Apply & Snapshot button.
Accepting atomically snapshots the 5MB state blob for a one-tap revert.

Guardrails & Safety



Persistent Memory

Stated limitations (e.g., “bad right shoulder”) persist in the Coach Profile. Every subsequent proposal respects this constraint.



Conservative by Design

If pain is mentioned in free text, the Coach will not diagnose. It programs conservatively and prompts professional consultation.



Offline Capable

If the AI times out or the CLI dies, the app never hangs. The Progression Engine continues offline without skipping a beat.

What Actually Leaves the Server?

Included - Minimized Payload (FR-10)

```
{  
  "routine_structure": "...",  
  "review_window": ["dates",  
    "exID", "RIR", "targets"],  
  "stalls": 2,  
  "e1RM_trend": "up",  
  "bodyweight": "[series]"  
}
```

Excluded - Stripped at Boundary (FR-11)

```
{  
  "passkeys": "...",  
  "ADMIN_UIDS": "...",  
  "display_names": "...",  
  "other_profiles": "...",  
  "push_subscriptions": "..."  
}
```

**Only the math and the effort leaves the box.
PII and accounts stay strictly local.**

The Coach Log (Auditability)

Timestamp: 2026-08-01T08:00:00Z

Trigger: [Scheduled Weekly]

Provider: claude-code CLI

Outcome: [Proposal Generated - 4 Changes]

User Decision: Accepted (2), Rejected (2)

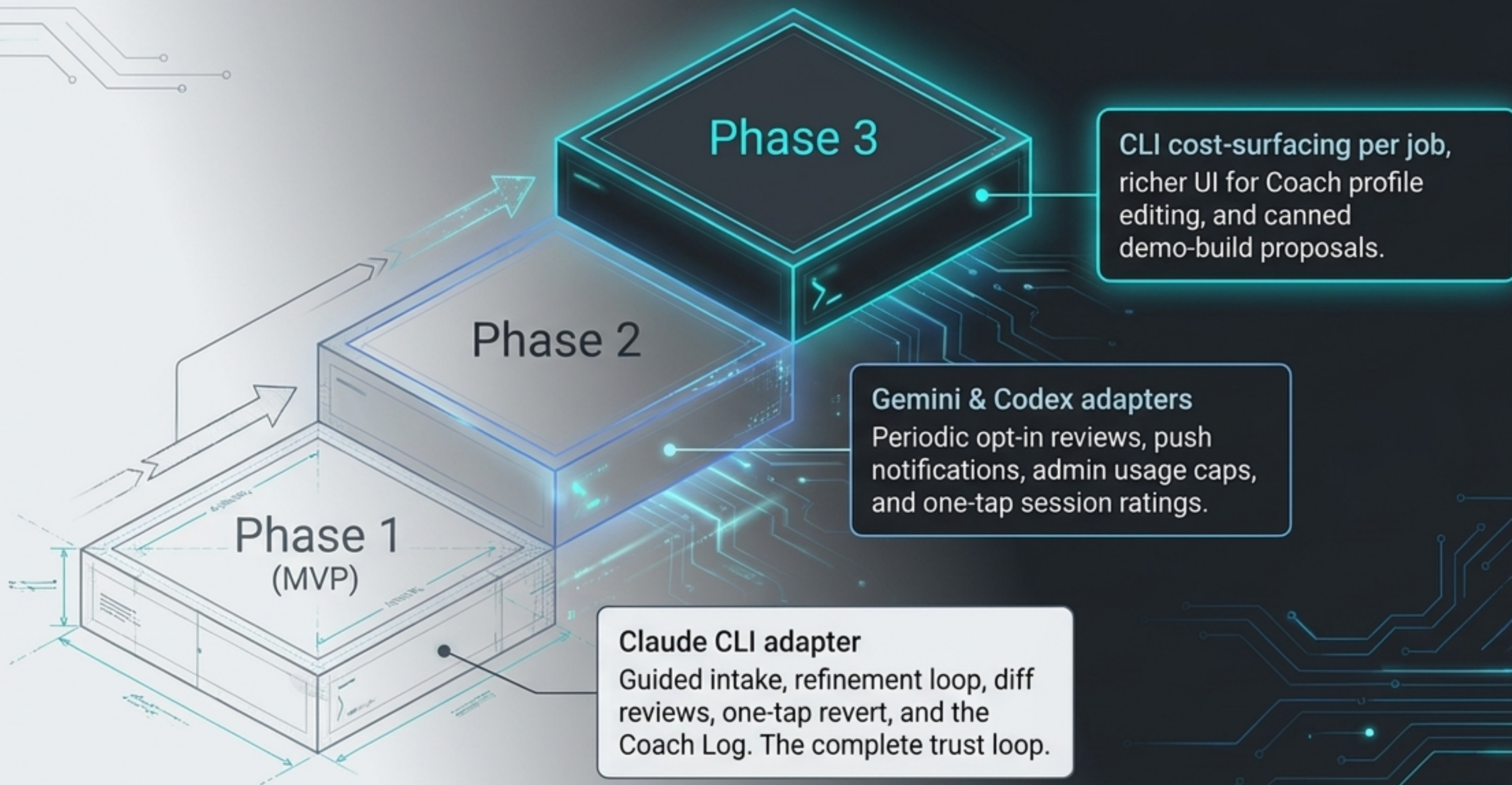
Every trigger, payload summary, and user decision is synced across devices, exportable in the full JSON backup, and fully auditable by the lifter.

Explicitly Out of Scope (For Now)

-  **No AI overriding tomorrow's per-session load.**
(The Engine owns the math).
-  **No live mid-workout chat or real-time set advice.**
(Keep sessions distraction-free).
-  **No form video analysis or biomechanical diagnostics.**
(Keep payloads minimized).
-  **No AI execution for browser-only Guest Mode.**
(Maintain the zero-telemetry promise).

We constrain the AI to plan design so the daily math remains deterministic.

Delivery Phasing



**The rigid math of
the progression
engine.**

openGym v1.3.0
**The smartest coach is the one
running on your own machine.**

**The dynamic
judgment of
local AI.**